# Sharp spectral gaps for the switch and Curveball chains on every graphical degree sequence

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**Abstract**

For every graphical degree vector on $n \geq 4$ labeled vertices with more than one realization, the lazy chain that proposes switches uniformly on four vertices has spectral gap at least $1/(6\binom{n}{4})$, and the chain that performs a uniform undirected Curveball trade on a uniformly chosen vertex pair has spectral gap at least $2/\binom{n}{2}$; both constants are the worst case over all such degree vectors, for every $n$. Consequently the total-variation mixing times at distance $1/4$ are at most $n^6/4+1$ for the switch chain and $n^4/16+1$ for the Curveball chain. Both gap bounds are kernel-checked in Lean, together with a total-variation bound for the switch chain whose prefactor is $\sqrt{M}/2$, where $M$ is the number of realizations, in place of the $\sqrt{M-1}/2$ used here.

## 1 Introduction

Let $V = \{1, \dots, n\}$, where $n \geq 4$, and let $d = (d_1, \dots, d_n)$ be a graphical degree vector. Write $\Omega_d$ for the set of simple undirected graphs on $V$ whose labeled degree vector is $d$, put $M = |\Omega_d|$, and let $\pi$ be the uniform probability measure on $\Omega_d$. A *switch* removes two disjoint edges and inserts a different perfect matching on their four vertices, provided both new edges are absent. A *Curveball trade* at a vertex pair $\{i, j\}$ redistributes the vertices adjacent to exactly one of $i$ and $j$ between these two endpoints, keeping both endpoint degrees and every other adjacency. Both moves preserve simplicity and every vertex degree.

We use the following precise discrete-time conventions. The switch chain $P$ holds with probability $1/2$. Otherwise it chooses a four-set uniformly and one of the six ordered pairs of distinct perfect matchings on that set uniformly. It removes the first matching and adds the second if all removed edges are present and all added edges absent; otherwise it holds. The Curveball chain $K$ chooses an unordered vertex pair uniformly and moves to a graph chosen uniformly among those obtainable from the current one by a Curveball trade at that pair, the current graph included; no additional holding probability is imposed. Time includes rejected proposals and holds for $P$, and counts trades for $K$. For probability measures on $\Omega_d$ and $Q \in \{P, K\}$, set

$$\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2} \sum_{G \in \Omega_d} |\mu(G) - \nu(G)|, \qquad t_{\mathrm{mix}}^{Q}(\varepsilon) = \min\Big\{ t \in \mathbb{Z}_{\geq 0} : \max_{G \in \Omega_d} \|Q^t(G, \cdot) - \pi\|_{\mathrm{TV}} \leq \varepsilon \Big\}.$$

For $M > 1$ write $\mathrm{gap}(Q) = 1 - \lambda_2(Q)$, where $\lambda_2(Q)$ is the second largest eigenvalue of $Q$, which is symmetric (Section 2). Here and throughout, logarithms are natural.

**Theorem 1.1.** *For every graphical $d$ on $n \geq 4$ vertices with $M > 1$,*

$$\operatorname{gap}(P) \geq \frac{1}{6\binom{n}{4}}, \qquad \operatorname{gap}(K) \geq \frac{2}{\binom{n}{2}}. \tag{1.1}$$

*For every $n \geq 4$, both constants are best possible universal constants, that is, the worst case over $d$: equality holds in both bounds for $d = (2, 2, 1, 1, 0, \ldots, 0)$.*

Equality holds whenever $M = 2$. Section 7 proves this, describes the extremal instances, and states what is not claimed for larger realization spaces.

**Corollary 1.2** (Mixing times)**.** *For $M > 1$ and $0 < \varepsilon \leq 1/2$, set $A_\varepsilon = \log(1/(2\varepsilon)) + \frac{1}{2}\log(M-1)$. Then*

$$t^P_{\mathrm{mix}}(\varepsilon) \leq \left\lceil 6\binom{n}{4} A_\varepsilon \right\rceil, \qquad t^K_{\mathrm{mix}}(\varepsilon) \leq \left\lceil \frac{\binom{n}{2}}{2} A_\varepsilon \right\rceil, \tag{1.2}$$

$$t^P_{\mathrm{mix}}(1/4) \leq n^6/4 + 1, \qquad t^K_{\mathrm{mix}}(1/4) \leq n^4/16 + 1. \tag{1.3}$$

*If $M = 1$, both mixing times are zero.*

No sharpness of these mixing bounds is asserted.

**History and significance.** Uniform sampling of graphs with prescribed degrees is a basic problem for degree-constrained network models. Kannan, Tetali and Vempala [14] proved rapid mixing of the switch chain for near-regular bipartite degree sequences and proposed the general rapid-mixing problem. Cooper, Dyer and Greenhill [5, 6] established polynomial mixing for regular undirected graphs. Greenhill and Sfragara [12] extended this to irregular undirected degrees under sparsity and degree conditions, and treated directed degrees under additional hypotheses including irreducibility. Erdős, Greenhill, Mezei, Miklós, Soltész and Soukup [9] proved rapid mixing on $P$-stable families of undirected, bipartite and directed degree sequences. These results impose restrictions on the degree families. Tikhomirov and Youssef [16] proved sharp Poincaré bounds for regular bipartite graphs in a specified degree range and sharp log-Sobolev bounds for fixed degree; their setting is distinct from arbitrary undirected degrees.

Verhelst [17] developed trade-based MCMC sampling for binary matrices with fixed marginals, including a uniform-stationary variant. Carstens, Berger and Strona [3] defined undirected Curveball trades and proved convergence to the uniform distribution. Their mixing-rate comparisons are experimental, and they left theoretical rapid mixing open. Carstens and Kleer [4] used the Johnson spectrum to compare switch and Curveball relaxation times for binary matrices, including irreducible spaces with forbidden entries. Their comparison transfers rapid mixing from the switch chain; it supplies no universal absolute gap, and the undirected comparison is left open in their preprint.

The closest universal results are those of OpenAI [15] and of Fu, Qin and Wang [11]. The manuscript [15] is machine-checked in Lean and has not been peer reviewed. It develops the pair-fiber framework used here, with the triple reduction and the disjoint-cell and slice Gram methods, and proves that $t^P_{\mathrm{mix}}(1/4) \leq 2n^8$ and that $P$ has spectral gap at least $[24n^2\binom{n}{4}]^{-1}$ [15, Theorem 1.1]. Fu, Qin and Wang prove sharp swap-chain and row-pair resampling bounds for all feasible binary matrix margins; their Theorem 5.1 also gives the three-row input used here. Arbitrary row-pair moves do not preserve symmetric zero-diagonal adjacency matrices, so their global matrix bound does not directly imply Theorem 1.1. Theorem 1.1 replaces the gap bound of [15, Theorem 1.1] by the best possible universal constant for the same chain, and gives the best possible universal constant for undirected Curveball.

**Proof strategy.** For an unordered vertex pair $a$, let $E_a$ average a function over the graphs reachable by a Curveball trade at $a$, put $h_a = I - E_a$ and $H = \sum_a h_a$, and let $L$ be the unnormalized Laplacian of the switch graph on $\Omega_d$. Then $I - P = L/(12\binom{n}{4})$ and $K = I - H/\binom{n}{2}$ (Section 2), so Theorem 1.1 reduces to $L \succeq H \succeq 2(I - \Pi)$, where $\succeq$ is the order of quadratic forms and $\Pi$ is projection onto constants. The comparison $H \preceq L$ is the classical Johnson comparison. The second inequality follows from $H^2 \succeq 2H$ and $\ker H = \mathbb{R}\mathbf{1}$. As in [15, Proposition 7.1], we expand

$$H^2 = \sum_{|T|=3} H_T^2 - (n-3)H + D,$$

where $H_T$ sums $h_a$ over the three pairs $a$ inside a vertex triple $T$, and $D$ is the ordered sum of the products $h_a h_b$ over disjoint pairs $a, b$. OpenAI proved $H_T^2 \succeq H_T$ and $D \succeq 0$, hence $H^2 \succeq H$ [15, Propositions 4.1 and 7.1, Corollary 6.3]. That intermediate estimate, with the constant kernel and the classical Johnson comparison, already gives $\mathrm{gap}(P) \geq 1/(12\binom{n}{4})$ and $\mathrm{gap}(K) \geq 1/\binom{n}{2}$, and, by the argument of Corollary 1.2, $t_{\mathrm{mix}}^P(1/4) \leq n^6/2 + 1$. In particular the inverse-fourth order follows from the ingredients of [15] plus the Johnson comparison; it is not the new part of this paper, and the improvement of Corollary 1.2 over those ingredients is the factor two.

The new step is the weighted estimate $D \succeq H$, which gives $H^2 \succeq (n-2)H - (n-3)H + H = 2H$. It yields the exact factor two and the Curveball constant $2/\binom{n}{2}$; Section 7 shows separately that both constants are attained. We prove $D \succeq H$ in two weighted steps. For each disjoint pair, the interaction $h_a h_b + h_b h_a$ is bounded below by twice a cross-square projection minus projections onto centered equality indicators, each weighted by the probability $p_a$ that two singleton labels of its fiber receive the same endpoint. On every slice, the Johnson energy minus the weighted equality projections still dominates the centered identity. Without the weights this slice inequality fails (Remark 6.2). The cell analysis follows [15, Proposition 5.1] up to its final square, and the slice analysis reuses the Gram decomposition of [15, Lemma 6.1]. New here are the weights $p_a$ with the completed square in the cell estimate, the exact second slice-moment identity, the weighted slice inequality $G - pS \succeq h$, and the assembly $D \succeq H$.

**Organization.** Section 2 defines the pair fibers and operators and identifies both chains with them. Section 3 proves the Johnson gap and $H \preceq L$. Section 4 proves the weighted inequality for one disjoint pair, and Section 5 the weighted slice inequality. Section 6 proves connectivity, $D \succeq H$ and $H^2 \succeq 2H$, and tracks the constants through Theorem 1.1 and Corollary 1.2. Section 7 proves equality whenever $M = 2$ and gives extremal instances. Section 8 describes the Lean verification, and Section 9 states open problems. Appendix A proves the three-row and graph-triple inputs. All proof ingredients are supplied in the paper.

## 2 Operators and pair fibers

This section fixes the operator conventions and expresses both chains through the switch Laplacian and the pair-fiber projections. All operators act on real $L^2(\pi)$. Write $A \succeq B$ when $A - B$ is self-adjoint and has nonnegative quadratic form. Let $\Pi f = \mathbb{E}_\pi f$ be projection onto constants.

Each distinct switch neighbor of a graph has transition probability $1/(12\binom{n}{4})$ under $P$: the four toggled edges determine both the four-set and the ordered pair of removed and added matchings, so this count has no proposal multiplicity. If $L$ is the unnormalized Laplacian of the graph of distinct switch neighbors, then

$$Lf(G) = \sum_{G' \sim G} (f(G) - f(G')), \qquad I - P = \frac{L}{12\binom{n}{4}}. \tag{2.1}$$

For an unordered vertex pair $a = \{i, j\}$, its *fiber* fixes all edges away from $a$, the edge $ij$, and each outside vertex's number of neighbors in $a$. The graphs in the fiber of $G$ are exactly those obtainable from $G$ by a Curveball trade at $a$. Let $E_a$ be conditional uniform averaging on this fiber and put

$$h_a = I - E_a, \qquad H = \sum_{a \in \binom{[n]}{2}} h_a, \qquad K = \frac{1}{\binom{n}{2}} \sum_a E_a = I - \frac{H}{\binom{n}{2}}. \tag{2.2}$$

This is the partition-based heat-bath construction of Dyer, Greenhill and Ullrich [8, Definition 1.1], so $E_a$ and $h_a$ are orthogonal projections. Thus $K$ chooses a uniform unordered pair and performs a uniform undirected Curveball trade [3], including the possibility of retaining the current assignment. No additional holding probability is imposed. Both kernels are symmetric and stochastic. Also $P$ has nonnegative spectrum by laziness, and $K$ does because it is an average of orthogonal projections. Section 3 describes each fiber as a uniform subset slice, and Section 6 combines the estimates for $H$ with (2.1) and (2.2).

## 3 Johnson comparison

This section describes each pair fiber as a uniform subset slice, bounds the pair projection $h_a$ by the exchange Laplacian of its fiber, and sums these bounds to obtain $H \preceq L$. The gap of the exchange Laplacian is a consequence of the classical Bernoulli–Laplace (Johnson) spectrum; see Diaconis and Shahshahani [7]. We give a self-contained local-to-global proof of the gap needed here. The identity $\sum_a J_a = 2L$ is that of OpenAI [15, Lemma 7.4], with the two-trade count of Carstens, Berger and Strona [3, Lemma 2]; the exact gap replaces the factor $n^2$ in the bound $h_a \preceq n^2 J_a$ of [15, Lemma 7.4]. Section 5 uses the gap on slices, and Section 6 uses $H \preceq L$.

On an $a = \{i, j\}$ fiber, let $N$ count outside vertices with exactly one neighbor in $a$, and let $c$ count those with two. Exactly $q = d_i - c - \mathbf{1}_{ij \in E}$ singleton labels must be assigned to $i$. Every $q$-subset of the $N$ labels is allowed, since all endpoint and outside degrees are preserved. There is one graph per subset, so the conditional law is uniform. Let $\tau_{kl}$ exchange the assignments of labels $k, l$, and put

$$J_a = \sum_{k<l} (I - \tau_{kl}).$$

Equal assignments give zero contributions. This is the unnormalized Johnson Laplacian, with each exchange neighbor counted once.

**Lemma 3.1** (Johnson gap)**.** *On a nontrivial uniform $q$-slice of $N$ labels, the gap of $J = \sum_{i<j}(I - \tau_{ij})$ is $N$. Consequently $h_a \preceq J_a/2$ and $H \preceq L$.*

*Proof.* Write $A_a = I - \tau_{ij}$ for a label pair $a = \{i, j\}$, so $A_a/2$ is an orthogonal projection and $A_a^2 = 2A_a$. Suppose first that $N \geq 3$. For each label triple $T$ set $J_T = \sum_{a \subset T,\, |a|=2} A_a$. Fixing the subset outside $T$ leaves either one assignment, when zero or three labels of $T$ are occupied, or three assignments, when one or two are occupied. On the latter orbit $J_T$ is the triangle Laplacian $3I - \mathbf{1}\mathbf{1}^{\mathsf{T}}$; on the former it is zero. Therefore

$$J_T^2 = 3J_T, \qquad \sum_{|T|=3} J_T = (N-2)J.$$

Swaps of disjoint label pairs commute, so $A_a A_b$ is positive semidefinite when $a \cap b = \varnothing$: it is four times the product of two commuting orthogonal projections. Define the ordered remainder

$$R_J = \sum_{\substack{a,b \in \binom{[N]}{2} \\ a \cap b = \varnothing}} A_a A_b \succeq 0.$$

In the sum of triple squares, each $A_a^2$ occurs $N-2$ times and each ordered intersecting pair of distinct label pairs occurs once. Expanding $J^2$ and using $A_a^2 = 2A_a$ gives

$$\begin{aligned} J^2 &= \sum_{|T|=3} J_T^2 - 2(N-3)J + R_J \\ &= 3(N-2)J - 2(N-3)J + R_J = NJ + R_J \succeq NJ. \end{aligned}$$

For $N = 2$ the only nontrivial slice has $q = 1$, and $J^2 = 2J = NJ$ directly. Thus every positive eigenvalue of the positive self-adjoint $J$ is at least $N$. Moreover,

$$\langle f, Jf\rangle = \tfrac{1}{2}\sum_{i<j}\|f - \tau_{ij}f\|^2.$$

Zero energy forces invariance under every label transposition. Such transpositions connect all $q$-subsets, so $\ker J = \mathbb{R}\mathbf{1}$. Finally, with $x_i$ the membership indicator of label $i$, direct summation gives $Jx_i = \sum_{j\neq i}(x_i - x_j) = Nx_i - q$. Since $1 \le q < N$, $x_i - q/N$ is a nonzero centered eigenfunction with eigenvalue $N$. The gap is therefore exactly $N$.

On a nontrivial fiber $N \ge 2$, so $h_a \preceq J_a/N \preceq J_a/2$; on one-state fibers both operators vanish. Every switch is realized by exactly two singleton exchanges: their indexing pairs are the two parts of the unique bipartition of its toggled four-cycle. For each indexing pair the outside labels are determined. Conversely, exchanging a label $k$ assigned to $i$ with a label $l$ assigned to $j$ toggles exactly the four edges $ik, jl, il, jk$, which is a valid switch, so every nontrivial singleton exchange is a switch. Thus, by off-diagonal entries and zero row sums,

$$\sum_a J_a = 2L, \qquad H \preceq \tfrac{1}{2}\sum_a J_a = L. \tag{3.1}$$

□

*Remark* 3.2. The Johnson proof has the same local-to-global shape as the argument for $H^2 \succeq 2H$: a triple square identity or inequality, a regrouping of interactions, and a constant-kernel argument. Disjoint label swaps commute, making $R_J$ positive directly. Disjoint graph-pair resamplings need not commute; Sections 4–6 control their interaction by the weighted estimates.

# 4 Weighted cell inequality

This section bounds the interaction $h_a h_b + h_b h_a$ of two disjoint pair resamplings from below. The proof follows the disjoint-pair proposition of OpenAI [15, Proposition 5.1] except for the final weighted square. The new ingredient is the fiber-constant weight $p_a$ on the error terms, which retains the identity $\alpha\beta = \rho^2$ in that square. Section 6 sums the lemma over disjoint pairs and pays for the error terms with Lemma 5.1.

On an $a$-fiber with parameters $(N, q)$ define the fiber-constant multiplier

$$p_a = \frac{q(q-1) + (N-q)(N-q-1)}{N(N-1)} \quad (N \ge 2), \qquad p_a = 0 \quad (N \le 1). \tag{4.1}$$

It is the probability that two distinct singleton labels have equal assignments. For disjoint $a, b$, let $T_{a,b}$ toggle the cross square between $a$ and $b$ when it is alternating, and fix the graph otherwise. It is a measure-preserving involution. Set $g_{a,b} = (I - T_{a,b})/2 = g_{b,a}$, an orthogonal projection. Let $R_{a,b}$ act on each $a$-fiber as projection onto the centered indicator that the two labels of $b$ have equal assignments; it is zero unless both are singletons, and also zero if that centered indicator vanishes. Consequently $p_a R_{a,b}$ is self-adjoint and positive semidefinite.

**Lemma 4.1** (Weighted disjoint-pair interaction)**.** *For disjoint vertex pairs $a, b$,*

$$h_a h_b + h_b h_a \succeq 2g_{a,b} - p_a R_{a,b} - p_b R_{b,a}. \tag{4.2}$$

*Proof.* Write $a = \{i, j\}$, $b = \{k, l\}$. Partition graphs into cells by fixing the graph outside these four vertices, the edges $ij, kl$, and each outside vertex's separate neighbor counts in $a$ and $b$. Both updates preserve these data. Let $A_0, B_0$ be the two outside singleton sets. After subtracting fixed double-neighbor and internal-edge contributions, let $q_0, q_1$ be the residual degrees of $i, k$. A cell is exactly the equally likely encodings $(F, U, W)$, where $F$ is a binary $2 \times 2$ cross matrix of fixed total $s$, $r, c$ are its first row and first column sums, and

$$U \subseteq A_0, \quad |U| = q_0 - r, \qquad W \subseteq B_0, \quad |W| = q_1 - c.$$

The total $s$ is fixed by the sum of the degrees in $a$. The displayed conditions enforce the first endpoint degrees; the second endpoint degrees then follow from the totals. All outside degrees are fixed as well. Even if $A_0, B_0$ overlap, their assignments use distinct edges and impose no additional constraint. This proves the claimed bijection and uniform law.

Within a cell the full $a$-fiber signature is $X = (c, W)$ and the full $b$-fiber signature is $Y = (r, U)$. Let $\mathcal{A}, \mathcal{B}$ be their function spaces, including constants, and $\mathcal{J} = (\mathcal{A} + \mathcal{B})^{\perp}$. The cross involution preserves $X, Y$, so $\operatorname{ran} g_{a,b} \subseteq \mathcal{J}$ and $P_{\mathcal{J}} \succeq g_{a,b}$.

**Product cases.** For $s = 0, 4$ the cross matrix is fixed. For $s = 1, 3$, its unique one or unique zero is determined by independent row and column choices and the subset weights factor. Thus $X, Y$ have a product law in these cases. For $s = 2$ the cross-matrix multiplicities are

$$(m_{rc})_{r,c=0}^{2} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \tag{4.3}$$

Set $\Delta_r = \{r \neq 1\}$ and $\Delta_c = \{c \neq 1\}$; they are disjoint. If either event is absent, one total is fixed and the remaining weights factor, again giving a product law. In every product case the conditional projections commute, their product is projection onto constants, and $h_a h_b = P_{\mathcal{J}}$. This proves (4.2) there.

**The rank-one case.** In the remaining case put $p_r = \Pr(\Delta_r) > 0$ and $p_c = \Pr(\Delta_c) > 0$. The occurrence of $\Delta_c$ ensures a feasible $U$ of size $q_0 - 1$, and that of $\Delta_r$ ensures a feasible $W$ of size $q_1 - 1$. Hence the central cross matrices are feasible, and $p_r + p_c < 1$. Define

$$u = \frac{\mathbf{1}_{\Delta_c} - p_c}{\sqrt{p_c(1 - p_c)}}, \quad v = \frac{\mathbf{1}_{\Delta_r} - p_r}{\sqrt{p_r(1 - p_r)}}, \quad \langle u, v \rangle = -\rho = -\sqrt{\frac{p_c p_r}{(1 - p_c)(1 - p_r)}}.$$

In particular $0 < \rho < 1$. Conditional on $X = (c, W)$, the weight of each feasible $(r, U)$ is proportional to $m_{rc}$, with normalizer $\sum_t \binom{|A_0|}{q_0 - t} m_{tc}$. By (4.3) this conditional law depends only on whether $c = 1$; the reverse statement holds for $r$. The centered correlation therefore has rank one: $E_a v = -\rho u$, $E_b u = -\rho v$. The centered parts of $\mathcal{A}$ orthogonal to $u$, and of $\mathcal{B}$ orthogonal to $v$, are orthogonal to the opposite space. They, constants, $\operatorname{span}(u, v)$, and $\mathcal{J}$ form an orthogonal invariant decomposition.

The interaction is zero on constants and those remaining row spaces, and is $2I$ on $\mathcal{J}$. On the plane write the component of a general $f$ as $xu + yv$. Its interaction quadratic form is

$$\Phi(f) = 2\langle h_a(xu + yv), h_b(xu + yv) \rangle = 2xy\rho(1 - \rho^2),$$

and its defect images are $y(v + \rho u)$ and $x(u + \rho v)$, with squared norms $y^2(1 - \rho^2)$ and $x^2(1 - \rho^2)$.

**The weighted square.** The line spanned by $v+\rho u$ lies in $\operatorname{ran} R_{a,b}$: it vanishes for $c \neq 1$, and on each $c=1$ fiber it is a multiple of the centered equal-assignment indicator. The analogous assertion holds for $u+\rho v$ and $R_{b,a}$. Because these lines lie in the plane, orthogonal components of the full $f$ do not affect its projection onto them. Therefore

$$\|R_{a,b}f\|^2 \geq y^2(1-\rho^2), \qquad \|R_{b,a}f\|^2 \geq x^2(1-\rho^2).$$

Every fiber supporting $R_{a,b}$ has $(N,q) = (|A_0|+2, q_0)$. Hence $p_a$ takes the value

$$\alpha := \Pr(\Delta_r \mid c=1) = \frac{p_r}{1-p_c}$$

on every $a$-fiber where $R_{a,b}$ is nonzero, and $p_b$ takes the value $\beta := p_c/(1-p_r)$ on every $b$-fiber where $R_{b,a}$ is nonzero. Indeed $c=1$ makes both labels of $b$ singletons in the $a$-fiber, and they have equal assignments exactly when $r \neq 1$. The uniform slice probability is precisely (4.1), independently of $W$. Now $\alpha\beta = \rho^2$, so

$$\begin{aligned}\Phi(f) + \langle f, p_a R_{a,b} f\rangle + \langle f, p_b R_{b,a} f\rangle &\geq (1-\rho^2)(2\rho xy + \alpha y^2 + \beta x^2) \\ &= (1-\rho^2)(\sqrt{\beta}x + \sqrt{\alpha}y)^2 \geq 0.\end{aligned}$$

The remaining contribution $2\|P_{\mathcal{J}} f\|^2$ is at least $2\|g_{a,b} f\|^2$. This proves the inequality on every cell and hence globally. □

## 5 Weighted slice inequality

This section concerns a uniform subset slice and has no graph interpretation. It compares the Johnson energy with the projections onto centered equality indicators, weighted by the equality probability $p$ of (4.1). The proof reuses the Gram decomposition and the first slice-moment computation of OpenAI [15, Lemma 6.1], which rest on Filmus's decomposition of the Johnson scheme [10, Lemmas 18 and 24]. The second slice-moment identity and the weighted inequality are new; without the weight $p$ the inequality fails (Remark 6.2). Section 6 applies the lemma on every pair fiber.

On a uniform $q$-slice of $N$ labels let $x_i$ be the membership indicators, $h = I - \Pi$, where here $\Pi$ denotes the slice average, $G = J/2$, and let $p$ be given by (4.1). Set

$$z_{ij} = \mathbf{1}_{x_i = x_j} - p, \qquad S = \sum_{i<j} Q_{ij},$$

where $Q_{ij}$ is projection onto $z_{ij}$, interpreted as zero if $z_{ij} = 0$.

**Lemma 5.1** (Weighted slice inequality)**.** *On every uniform slice, $G - pS \succeq h$.*

*Proof.* One-state slices have $G = S = h = 0$. For $N = 2, q = 1$, $p = S = 0$ and $G = h$. Otherwise $N \geq 3$, $1 \leq q < N$, and $0 < p < 1$. Put $r = N - q$ and $m_N = \binom{N}{2}$. Map the Euclidean space of pair coefficients into the slice space by

$$Bc = \sum_{i<j} c_{ij} \frac{z_{ij}}{\sqrt{p(1-p)}}.$$

Then $S = BB^*$. The coefficient space decomposes orthogonally as

$$C_0 = \mathbb{R}\mathbf{1}, \quad C_1 = \{c_{ij} = u_i + u_j : \sum_i u_i = 0\}, \quad C_2 = \{c : \sum_{j \neq i} c_{ij} = 0 \text{ for every } i\}.$$

Here $c_{ji} = c_{ij}$. The map $u \mapsto (u_i + u_j)$ is injective for $N \geq 3$; the orthogonal complement of its whole image is exactly $C_2$. The dimensions are $1, N-1, m_N - N$, so $C_2 = 0$ if $N = 3$.

**The Gram scalars.** The number of equal pairs is fixed; hence $BC_0 = 0$. By label symmetry the entries of $B^*B$ depend only on the intersection size of the indexing pairs. For a fixed pair $ij$, the sums of $c_{kl}$ over other pairs meeting it and over disjoint pairs are respectively $(N-4)c_{ij}, -(N-3)c_{ij}$ on $C_1$, and $-2c_{ij}, c_{ij}$ on $C_2$. These follow by summing the row sums (which are $(N-2)u_i$ on $C_1$) and using the zero total sum. Thus $B^*B$ acts as scalars $\mu_1, \mu_2$. Its diagonal entries are one. The identities

$$\sum_{i<j}(u_i+u_j)z_{ij} = (2q-N)\sum_i u_i x_i,$$
$$\operatorname{Var}\Big(\sum_i u_i x_i\Big) = \frac{qr}{N(N-1)}\sum_i u_i^2, \qquad \sum_{i<j}(u_i+u_j)^2 = (N-2)\sum_i u_i^2$$

follow from $\sum_i u_i = 0$, $\sum_i x_i = q$, and $\mathbb{E}x_i = q/N$, $\mathbb{E}x_i x_j = q(q-1)/(N(N-1))$. Since $1-p = 2qr/(N(N-1))$, they give the first formula below; the trace identity $(N-1)\mu_1 + (m_N - N)\mu_2 = m_N$ gives the second:

$$p\mu_1 = \frac{(N-2q)^2}{2(N-2)}, \qquad p\mu_2 = \frac{2(q-1)(r-1)}{(N-2)(N-3)} \quad (N \geq 4).$$

**Comparison on the images.** The mutually orthogonal images $U_i = BC_i$ exhaust $\operatorname{ran} S$, and $S$ acts there by $\mu_i$. Direct transposition summation gives

$$Gx_i = (Nx_i - q)/2, \qquad G(x_i x_j) = (N-1)x_i x_j - (q-1)(x_i + x_j)/2.$$

The image $U_1$ is linear. For $C_2$, expansion of $z_{ij} = 1 - p - x_i - x_j + 2x_i x_j$ cancels the constant and linear coefficients; its row sums also cancel the lower terms in the second displayed formula. Consequently $G$ acts by $N/2$ on $U_1$ and by $N-1$ on $U_2$. Zero images require no eigenvalue assertion. On the nonzero images the eigenvalues of $G - pS - h$ are nonnegative: on $U_1$,

$$\frac{N}{2} - p\mu_1 - 1 = \frac{2(q-1)(r-1)}{N-2} \geq 0;$$

on $U_2$, using $(q-1)(r-1) \leq (N-2)^2/4$,

$$0 \leq p\mu_2 \leq \frac{N-2}{2(N-3)} \leq 1, \qquad N - 1 - p\mu_2 - 1 \geq N - 3 > 0.$$

On the orthogonal complement of constants and $U_1 \oplus U_2$, $S = 0$, while Lemma 3.1 gives $G \succeq (N/2)I \succeq I$. These subspaces are invariant under the self-adjoint $G$; constants are killed by all the operators. This completes the proof, including all boundary cases. □

## 6 Assembly

This section sums the weighted cell and slice inequalities into $D \succeq H$ and $H^2 \succeq 2H$, identifies the kernel of $H$, and tracks the constants through Theorem 1.1 and Corollary 1.2. The triple estimate is Lemma A.4, proved in Appendix A.

### 6.1 The global operator inequality

For a vertex triple $T$ let $H_T = \sum_{a \subset T,\, |a|=2} h_a$, and define

$$D = \sum_{\substack{a,b \in \binom{[n]}{2} \\ a \cap b = \varnothing}} h_a h_b.$$

This is an ordered sum and is self-adjoint; no commutation is assumed. Using $h_a^2 = h_a$, each diagonal term occurs $n-2$ times in the triple sum, and each ordered pair of distinct intersecting pairs occurs once. Thus we obtain the expansion of OpenAI [15, Proposition 7.1], which follows Caputo's squared-generator method [1, Section 1.1, Equations (1.13)–(1.15)]:

$$H^2 = \sum_{|T|=3} H_T^2 - (n-3)H + D, \qquad \sum_{|T|=3} H_T = (n-2)H. \tag{6.1}$$

**Proposition 6.1** (Disjoint and global estimates)**.** *The disjoint-pair sum satisfies $D \succeq H$, and consequently $H^2 \succeq 2H$.*

*Proof.* On an $a$-fiber, pairs $b$ containing a nonsingleton label contribute zero to both $g_{a,b}$ and $R_{a,b}$. The other pairs index exactly the singleton-label transpositions and equality lines. Therefore

$$G_a := \sum_{b \cap a = \varnothing} g_{a,b} = J_a/2, \qquad S_a := \sum_{b \cap a = \varnothing} R_{a,b} = S \text{ on that fiber.}$$

Sum Lemma 4.1 over *unordered* disjoint $\{a, b\}$. Each such pair contributes two copies of $g_{a,b}$, giving exactly $\sum_a G_a$, and each directed error occurs once. Lemma 5.1, averaged over fibers, now gives

$$D \succeq \sum_a (G_a - p_a S_a) \succeq \sum_a h_a = H.$$

Using the triple estimate of Lemma A.4 in (6.1) yields

$$H^2 \succeq (n-2)H - (n-3)H + D \succeq 2H.$$

□

*Remark* 6.2. The weights cannot be removed from Lemma 5.1: on $q = 1$ slices with $N \geq 3$, $G = S$, so $G - S \succeq h$ is false, whereas $G - pS = h$. Retaining $\alpha\beta = \rho^2$ in the cell square leaves the slice energy needed for the strengthened global bound.

### 6.2 Connectivity and the kernel

The following lemma is the connectivity lemma and kernel corollary of OpenAI [15, Lemma 7.2, Corollary 7.3], whose connectivity argument is Havel's switching argument [13, Theorem 3]. We include the proof.

**Lemma 6.3** (Connectivity and kernel)**.** *Switches connect $\Omega_d$, and* $\ker H = \mathbb{R}\mathbf{1}$.

*Proof.* Let $R$ be the current set of unfrozen vertices, initially $V$, and write $d_w^R$ for the degree of $w$ in the graph induced on $R$ and $N_R(w)$ for its neighborhood there. Switches inside $R$ leave every residual degree $d_w^R$ unchanged, and hence also the prescription below. Choose the first vertex $v$ of $R$ by label and prescribe its neighborhood in $R$ to be the $d_v^R$ vertices of $R \setminus \{v\}$ of largest residual

degree, breaking ties by label. If $i$ is an undesired current neighbor and $j$ a prescribed missing neighbor, then $d_j^R \geq d_i^R$. The sets $A = N_R(j) \setminus \{i\}$ and $B = N_R(i) \setminus \{j\}$ satisfy $|A| \geq |B|$ and $v \in B \setminus A$. Hence some $k \in A \setminus B$ exists. These four vertices are distinct and $\{vi, jk\} \to \{vj, ik\}$ is a valid switch, gaining a prescribed neighbor without losing another. Repeat, then freeze $v$ and recurse. The resulting residual degree vector depends only on the preceding vector and the fixed prescription. Every realization therefore reaches the same graph. With fewer than four unfrozen vertices a discrepancy would still force four distinct vertices, so none occurs.

Since the $h_a$ are orthogonal projections, $\langle f, Hf\rangle = \sum_a \|h_a f\|^2$. Zero energy forces $f$ to be constant on every fiber and hence along every switch, by the multiplicity count behind (3.1). Connectivity proves the kernel statement, and constants have zero energy because $h_a\mathbf{1} = 0$. □

**Corollary 6.4** (Operator comparison)**.** *The operators satisfy* $L \succeq H \succeq 2(I - \Pi)$.

*Proof.* By Proposition 6.1, the positive self-adjoint $H$ has each eigenvalue either zero or at least two. Lemmas 6.3 and 3.1 imply $L \succeq H \succeq 2(I - \Pi)$. □

### 6.3 The prescribed chains

*Proof of Theorem 1.1.* By Corollary 6.4 and (2.1),

$$I - P = \frac{L}{12\binom{n}{4}} \succeq \frac{2(I - \Pi)}{12\binom{n}{4}} = \frac{I - \Pi}{6\binom{n}{4}},$$

and by (2.2), $I - K = H/\binom{n}{2} \succeq 2(I - \Pi)/\binom{n}{2}$. Both kernels are symmetric and fix constants, so these are the two bounds in (1.1). By Proposition 7.2, $d = (2, 2, 1, 1, 0, \ldots, 0)$ has $M = 2$, so Proposition 7.1 gives equality in both bounds. □

*Proof of Corollary 1.2.* The case $M = 1$ is immediate: the unique graph carries all of $\pi$. For $M > 1$ and either kernel, the centered density of a point mass has squared norm $M - 1$. Reversibility and the nonnegative spectrum give, for a valid gap lower bound $\gamma_0$ and every starting graph,

$$\|Q^t(G, \cdot) - \pi\|_{\mathrm{TV}} \leq \tfrac{1}{2}\sqrt{M - 1}(1 - \gamma_0)^t \leq \tfrac{1}{2}\sqrt{M - 1}e^{-\gamma_0 t}. \tag{6.2}$$

Substitute $\gamma_0 = 1/(6\binom{n}{4})$ or $2/\binom{n}{2}$ to obtain (1.2). Since $M \leq 2^{\binom{n}{2}}$,

$$A_{1/4} \leq \left(1 + \frac{n(n-1)}{4}\right)\log 2 \leq n^2, \qquad 6\binom{n}{4} \leq \frac{n^4}{4}.$$

For the second coarse constant, $n \geq 4$ gives

$$\frac{\binom{n}{2}}{2}A_{1/4} \leq \left(\frac{n^2}{4} + \frac{n^4}{16}\right)\log 2 = \frac{n^4}{16}(1 + 4/n^2)\log 2 \leq \frac{n^4}{16}.$$

Here $(1 + 4/n^2)\log 2 \leq (5/4)\log 2 < 1$, because $\log 2 = \int_1^2 dx/x < 3/4$ by the trapezoidal bound for the strictly convex function $1/x$. Finally $\lceil x \rceil \leq x + 1$ proves (1.3). □

## 7 Best possible constants

This section proves that the constants of Theorem 1.1 are best possible universal constants for every $n \geq 4$, that is, the worst case over $d$. Equality holds for every degree vector with exactly two realizations, and such vectors exist for every $n \geq 4$.

**Proposition 7.1** (Two realizations)**.** *If $M = 2$, equality holds in both bounds of (1.1).*

*Proof.* By Lemma 6.3 the two realizations are switch neighbors, so $L$ is the Laplacian of one edge, with gap two. Within a pair fiber each reassigned label toggles two edges, so two graphs of one fiber that differ in four edges differ by one singleton exchange. Hence a pair fiber contains both realizations only when the pair is one of the two parts of the bipartition of the toggled four-cycle, by the multiplicity count in the proof of Lemma 3.1. Every other pair fiber has one state, so exactly two $h_a$ are nonzero, and each equals $I - \Pi$ on the two-state space. Thus $H = 2(I - \Pi)$, and (2.1)–(2.2) give $I - P = (I - \Pi)/(6\binom{n}{4})$ and $I - K = 2(I - \Pi)/\binom{n}{2}$, which is equality in both parts of (1.1). □

**Proposition 7.2** (Extremal instances)**.** *For every $n \geq 4$, the degree vector $d = (2, 2, 1, 1, 0, \ldots, 0)$ has $M = 2$. The degree vector $(3, 3, 2, 1, 1)$ has $M = 2$ and neither an isolated nor a dominating vertex.*

*Proof.* Take $d = (2, 2, 1, 1, 0, \ldots, 0)$. The edge 12 must be present: otherwise both vertices $1, 2$ would require both $3, 4$, contradicting their degrees. The only realizations are

$$\{12, 13, 24\}, \qquad \{12, 14, 23\}.$$

They differ by one switch, and the two nontrivial pair fibers are indexed by $\{1, 2\}$ and $\{3, 4\}$; each is the whole two-state space.

Take $d = (3, 3, 2, 1, 1)$. The edge 12 must be present: otherwise both vertices $1, 2$ would be adjacent to all of $3, 4, 5$, giving vertex 4 degree two. The four remaining edges at 1 and 2 end in $\{3, 4, 5\}$, whose degree sum is four, so no edge lies inside $\{3, 4, 5\}$ and vertex 3 is adjacent to both 1 and 2. Each of $1, 2$ then has one neighbor in $\{4, 5\}$, and these neighbors are distinct because vertices $4, 5$ have degree one. The only realizations are therefore

$$\{12, 13, 23, 14, 25\}, \qquad \{12, 13, 23, 15, 24\}.$$

No degree is zero, and none equals $n - 1 = 4$. □

The instance $(2, 2, 1, 1, 0, \ldots, 0)$ is a two-element realization space with $n - 4$ isolated vertices appended. Nothing is claimed here about the best constants over degree vectors whose realization spaces are larger; whether equality or near-equality occurs for large $M$ is open (Section 9). For regular bipartite graphs in a specified degree range, Tikhomirov and Youssef [16] proved sharp Poincaré bounds for the switch chain, and sharp log-Sobolev bounds for fixed degree.

## 8 Formal verification

The two gap lower bounds of Theorem 1.1 and a weaker total-variation bound for $P$ are checked in Lean 4 with Mathlib. The switch statements use the definitions of the Lean formalization accompanying [15] at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, unchanged. The toolchain pins are as follows.

| | |
|---|---|
| Lean | `leanprover/lean4:v4.34.1` |
| Mathlib | `d13f23b723b8a846827a245b89c10fc7d3f11612` |

**Switch contracts.** The file `Statement.lean` imports only the pinned definitions and states two contracts. Both assume $n \geq 4$, a degree vector $d$ with $d_v \leq n-1$ for every $v$, $d$ graphical, and $M > 1$, where $\Omega_d$ is the pinned state space `GraphState` $n$ $d$ and $M$ is `stateCount` $n$ $d$. Let $P$ be the pinned `switchKernel`, $\mathrm{Var}_\pi$ the pinned `uniformVariance`, and $\mathcal{E}_P(f) = \langle f, (I-P)f \rangle$ the pinned `dirichletEnergy`. The contracts read

$$\frac{1}{6\binom{n}{4}} \mathrm{Var}_\pi(f) \leq \mathcal{E}_P(f) \quad \text{for every } f \colon \Omega_d \to \mathbb{R},$$

$$\|P^t(G,\cdot) - \pi\|_{\mathrm{TV}} \leq \frac{\sqrt{M}}{2} \exp\Big(-\frac{t}{6\binom{n}{4}}\Big) \quad \text{for every } t \in \mathbb{Z}_{\geq 0} \text{ and } G \in \Omega_d.$$

The pinned gap predicate is this Poincaré form: $\gamma$ times the variance is at most the Dirichlet energy for every $f$. For a reversible kernel the largest such $\gamma$ is $1-\lambda_2$, so the first contract is the switch bound in (1.1). The second contract has prefactor $\sqrt{M}/2$, weaker than the $\sqrt{M-1}/2$ of (6.2). The theorems `SharpSwitchGap.switch_chain_gap` and `SharpSwitchGap.switch_chain_tv_bound` prove the two statements, the latter by the general gap-to-total-variation argument of the upstream formalization. The theorems `SharpSwitchGap.Baseline.sharp_gap` and `SharpSwitchGap.Baseline.sharp_tv_bound` check these proofs against the two contracts. A synchronization script verifies the Git blob hash of every copied upstream file and matches the definitions and upstream theorem types against the pinned upstream statements.

**Curveball contract.** The project defines the Curveball kernel `Curveball.kernel` on the pinned state space: it chooses a uniform unordered pair and then a uniform state among the pinned `fiberStates` of that pair, the current state included. Both identities in (2.2) are proved by the theorems `kernel_apply` and `operator_average`; further theorems prove that the kernel is nonnegative, stochastic and symmetric, and that its operator is positive semidefinite. With $\mathcal{E}_K(f) = \langle f, (I-K)f \rangle$, the theorem `Curveball.gap` proves, for $n \geq 4$ and graphical $d$,

$$\frac{2}{\binom{n}{2}} \mathrm{Var}_\pi(f) \leq \mathcal{E}_K(f) \quad \text{for every } f \colon \Omega_d \to \mathbb{R}.$$

The Curveball kernel and its energy and gap predicates are defined in this project on the pinned graph space rather than taken from upstream, so this contract rests on a larger trusted base than the switch contracts.

**Checked argument.** Table 1 lists the intermediate results of the checked argument. The triple estimate $H_T^2 \succeq H_T$ enters as the pinned upstream formal result; the proofs in Appendix A are not separately formalized. The audited endpoints depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

| Statement | Lean declaration |
|---|---|
| $H \preceq L$ (Lemma 3.1) | `SharpSwitchGap.SharpComparison.generator_le_laplacian` |
| Lemma 5.1 | `SharpSwitchGap.WeightedSlice.weighted_centered` |
| Lemma 4.1, summed over cells | `SharpSwitchGap.WeightedCell.actual_named_interaction` |
| $D \succeq H$ | `SharpSwitchGap.SharpAssembly.disjoint_ge_generator` |
| $H^2 \succeq 2H$ | `SharpSwitchGap.SharpAssembly.generator_square` |
| $H \succeq 2(I - \Pi)$ | `SharpSwitchGap.SharpAssembly.centered_poincare` |

Table 1: Intermediate results and the Lean declarations that check them.

**Reproduction.** With the Lean toolchain available, run the following from the `lean` directory of the repository at https://github.com/sevenevesai/SharpSwitchGap.

```
python verify_vendor.py
lake exe cache get
lake build
lake env lean SharpSwitchGap/B5Checks.lean
lake env lean SharpSwitchGap/B4Checks.lean
```

The first command checks the vendored upstream files against the Git blob hashes of the pinned commit. The two audits print the axiom dependencies of the Curveball and switch endpoints and of the two contract checks. Success means exit code zero, no `sorry` diagnostic, and only the three axioms above in every printed dependency list.

**Not formalized.** The following results of the paper are not formalized:

- the equality cases, Propositions 7.1 and 7.2;
- the exact Johnson gap equality in Lemma 3.1; the lower bound is checked;
- the $\sqrt{M-1}/2$ prefactor in (6.2); the checked switch contract uses $\sqrt{M}/2$ with the same gap constant;
- a total-variation estimate for the Curveball chain $K$;
- the mixing-time bounds of Corollary 1.2.

# 9 Open problems

In the normalizations of Section 1, the worst-case relaxation times have orders $n^4$ for $P$ and $n^2$ for $K$, by Theorem 1.1. At fixed error, Corollary 1.2 gives only $O(n^6)$ and $O(n^4)$ mixing-time bounds, respectively. The loss comes from the $\log M$ factor in the gap-to-total-variation estimate. A log-Sobolev or entropy-contraction analogue of the pair-fiber argument strong enough to avoid this factor is not proved here.

Directed degree sequences require a separate extension. A formulation must specify an irreducible chain, since ordinary directed switches need not connect every realization space; the directed results in [12, 9, 4] have their own hypotheses. The weighted interaction argument is not extended here to the forbidden diagonal of a directed adjacency matrix.

The known extremal instances have two realizations (Proposition 7.1). Whether equality or near-equality holds in either bound of (1.1) for some degree vectors with large $M$ is not known. How much larger the gaps are for non-extremal degree sequences, or for typical sequences under a specified random model, is not quantified here.

# A Three-row and graph-triple estimates

This appendix proves the triple estimate $H_T^2 \succeq H_T$ used in Proposition 6.1. The three-row variance inequality, Lemma A.3, is the conditional-projection inequality of Fu, Qin and Wang [11, Theorem 5.1], in the equivalent form (A.1). We reproduce the alternative proof of OpenAI [15, Section 3]. Its fixed-block step, Lemma A.1, uses transposition eigenspaces, a strategy that [15,

Section 3] credits to the symmetric-group method of Carlen, Lieb and Loss [2, Section 2]. Its count step, Lemma A.2, uses an eigenvector increment argument with the adjacent-count coupling of Fu, Qin and Wang [11, Lemma 5.2]; the count law corresponds to their Lemmas 5.2–5.4. The transfer to graph triples, Lemma A.4, uses the triple encoding of OpenAI [15, Proposition 4.1]. All steps are included.

### A.1 A variance formulation

For three coordinates on any finite probability space let $V_i$ be the centered functions of coordinate $i$, and put the sum of marginal squared norms on $V_1 \oplus V_2 \oplus V_3$. For $A(g_1, g_2, g_3) = g_1 + g_2 + g_3$ the adjoint on centered joint functions is $A^* f = (P_1 f, P_2 f, P_3 f)$, where $P_i$ is conditional expectation onto that coordinate. This follows by testing inner products. Therefore the assertions

$$\operatorname{Var}\Big(\sum_{i=1}^{3} g_i\Big) \le 2 \sum_{i=1}^{3} \operatorname{Var}(g_i), \qquad \sum_{i=1}^{3} P_i \preceq 2I \text{ on } \mathbf{1}^{\perp} \tag{A.1}$$

are equivalent, since $AA^* = \sum_i P_i$ and both assertions say $\|A\|^2 \le 2$.

### A.2 Fixed block sizes

**Lemma A.1** (Fixed block sizes)**.** *Take independent uniform ordered partitions of labeled sets $U, W$ into three blocks of prescribed sizes, and let the row coordinates be $(U_i, W_i)$. Then (A.1) holds for these coordinates.*

*Proof.* Let $J$ be the sum of $I - T_\tau$ over label transpositions within either set. These are self-adjoint involutions, and $\langle f, Jf \rangle = \frac{1}{2} \sum_\tau \|f - T_\tau f\|^2$. The two permutation groups act transitively on the partition pairs, so $\ker J = \mathbb{R}\mathbf{1}$, including the degenerate one-state case.

Every transposition preserves each row-function space: it takes $U_i$ or $W_i$ to its image under the known label permutation, without needing the other blocks. Thus $J$ preserves these spaces. In finite dimension its spectral projectors are polynomials in $J$ (interpolate on the distinct eigenvalues), so they also preserve the row spaces. Center each $g_i$ and project it into each positive eigenspace, writing $g_{i,\lambda}$. At a given state any transposition changes at most two block tuples; hence at most two of the three differences $g_{i,\lambda} - T_\tau g_{i,\lambda}$ are nonzero. The inequality $(a+b)^2 \le 2(a^2 + b^2)$ gives

$$\lambda \Big\| \sum_i g_{i,\lambda} \Big\|^2 \le 2\lambda \sum_i \|g_{i,\lambda}\|^2.$$

Cancel $\lambda > 0$ and sum over orthogonal eigenspaces to obtain (A.1). The invariance of the row spaces is what permits passing from the transposition energy estimate to this variance estimate. □

### A.3 Random block sizes

**Lemma A.2** (Random block sizes)**.** *For nonnegative integers $m$ and $\alpha_1, \alpha_2, \alpha_3$ let $w_i(t) = 1/(t!(t+\alpha_i)!)$, and condition the product weights $\prod_i w_i(x_i)$ on $x_1 + x_2 + x_3 = m$. Then (A.1) holds for the coordinates $x_i$.*

*Proof.* All compositions of $m$ have positive probability. First consider two coordinates of total $s$, with $p_s(y) \propto w_j(y) w_k(s-y)$, $0 \le y \le s$. On $0 \le y < s$,

$$\frac{p_s(y)}{p_{s-1}(y)} = \frac{c_s}{(s-y)(s-y+\alpha_k)}, \qquad c_s > 0,$$

which increases with $y$. Reweighting by an increasing positive likelihood ratio increases expectations of increasing functions: for increasing $a, b$ and independent $Y, Y'$ with the old law,

$$2\operatorname{Cov}(a(Y), b(Y)) = \mathbb{E}[(a(Y) - a(Y'))(b(Y) - b(Y'))] \geq 0.$$

The extra mass at the new largest value $s$ preserves this stochastic order, so $Y_{s-1} \leq_{\mathrm{st}} Y_s$. Apply the same argument to $Z_s = s - Y_s$ to obtain $Y_s \leq_{\mathrm{st}} Y_{s-1} + 1$. Common-uniform quantiles simultaneously couple these variables as $Y_{s-1} \leq Y_s \leq Y_{s-1} + 1$. Thus increasing the total by one increases exactly one of the two integer coordinates by one.

Use the addition operator $A$ above for the centered marginal spaces. A nonzero eigenvector $b$ of $A^*A$ at eigenvalue $\lambda$ satisfies

$$\lambda b_i(t) = b_i(t) + \sum_{j \neq i} \mathbb{E}[b_j(x_j) \mid x_i = t], \qquad 0 \leq t \leq m.$$

If $m > 0$, set $\eta = \max_{i,t<m} |b_i(t+1) - b_i(t)| > 0$; otherwise centering would make $b$ zero. Adjacent values of $x_i$ change the other total by one. The coupling just proved changes exactly one of those coordinates by one, so the difference of their summed conditional expectations has absolute value at most $\eta$. Subtracting the eigenvector equations gives

$$|\lambda - 1|\,|b_i(t+1) - b_i(t)| \leq \eta.$$

At a maximizing increment, $|\lambda - 1| \leq 1$ and hence $\lambda \leq 2$. Apply this to a largest eigenvalue of the positive $A^*A$ to prove (A.1). If $m = 0$, every variance is zero. □

### A.4 All binary three-row margins

**Lemma A.3** (Three-row variance inequality)**.** *Take the uniform binary matrices with any feasible three-row margins. Then* (A.1) *holds for the three row coordinates.*

*Proof.* Delete columns of sum zero or three and subtract the sum-three count from the row totals. Let $U$ be the sum-one columns and $W$ the sum-two columns. Write $U_i$ for the unique-one locations in row $i$, $W_i$ for the unique-zero locations, and $r_i$ for the adjusted row total. Each row is equivalent to $(U_i, W_i)$, and the whole matrix to the two ordered partitions. With $k_i = |W_i|$ and $\delta_i = r_i - |W|$, one has $|U_i| = k_i + \delta_i$. At fixed $k$ there are exactly

$$\frac{|U|!\,|W|!}{\prod_i k_i!(k_i + \delta_i)!}$$

matrices, and the two uniform partition laws are independent. All these partitions are feasible by their unique-one and unique-zero interpretation.

Set $l_i = \max(0, -\delta_i)$, $x_i = k_i - l_i$, $\alpha_i = |\delta_i|$, and $m = |W| - \sum_i l_i$. Nonemptiness gives $m \geq 0$. The feasible $x$ are all nonnegative compositions of $m$; block upper bounds follow from their nonnegative sizes and fixed totals. Also $k_i!(k_i + \delta_i)! = x_i!(x_i + \alpha_i)!$, so the count law is the one of Lemma A.2. For a row function $f_i$, the conditional mean $b_i(k_i) = \mathbb{E}[f_i \mid k]$ depends only on $k_i$, because its marginal blocks are independent uniform subsets of sizes $k_i + \delta_i, k_i$. Total variance, Lemma A.1, and then Lemma A.2 give

$$\begin{aligned}\operatorname{Var}\Big(\sum_i f_i\Big) &= \mathbb{E}\operatorname{Var}\Big(\sum_i f_i \,\Big|\, k\Big) + \operatorname{Var}\Big(\sum_i b_i\Big) \\ &\leq 2\sum_i \mathbb{E}\operatorname{Var}(f_i \mid k) + 2\sum_i \operatorname{Var}(b_i) = 2\sum_i \operatorname{Var}(f_i).\end{aligned}$$

This proves (A.1) for every feasible three-row margin. □

### A.5 Encoding graph triples

Fix $T = \{i, j, k\}$, the graph outside $T$, and each outside vertex's number of neighbors in $T$. These data define cells preserved by pair updates inside $T$. The degree sum in $T$ minus the sum of outside counts determines twice the internal edge count $e$. Encode the outside incidences by three rows, and augment them as in Table 2, where $s_i$ is the internal degree of vertex $i$.

| $e$ | Internal graph | Extra column | Prescribed row sum |
|---|---|---|---|
| 0 | empty | none | $d_i$ |
| 1 | edge between marked vertices | $s_i$, sum 2 | $d_i$ |
| 2 | path centered at marked vertex | $s_i - 1$, sum 1 | $d_i - 1$ |
| 3 | triangle | none | $d_i - 2$ |

Table 2: Augmentation of the three-row encoding of a graph triple by its internal edge count $e$.

The extra column is a separate label and uniquely reconstructs the internal graph. Conversely every binary matrix of these margins reconstructs a graph in the cell: the row sums impose the three degrees, and the column sums and fixed outside graph impose all other degrees. There is no extra symmetry or structural-zero restriction. This is a bijection preserving uniform measure.

An augmented row $S_i$ fixes the external neighbors of $i$, hence its internal degree $s_i$ by subtraction from $d_i$; its extra entry adds no restriction. Together with the cell data it determines the full pair-fiber signature for $\{j, k\}$: all edges away from that pair, its internal edge $jk$ with indicator $e - s_i$, the other outside pair counts obtained by subtracting incidences to $i$, and the pair count $s_i$ of vertex $i$ itself. Conversely the signature determines the augmented row. In particular $s_i = 1$ fixes a count, not a choice of $j$ or $k$. Both conditionings average exactly the same sets, so $E_{\{j,k\}} = P_i = \mathbb{E}[\cdot \mid S_i]$ on the cell.

**Lemma A.4** (Graph triples)**.** *For every vertex triple $T$, $H_T^2 \succeq H_T$.*

*Proof.* Lemma A.3 and the adjoint formulation (A.1) give $\sum_i P_i \preceq 2I$ on centered cell functions. Thus $H_T = 3I - \sum_i P_i \succeq I$ there, while $H_T$ kills cell constants. Self-adjointness makes its eigenvalues zero or at least one, and hence $H_T^2 \succeq H_T$. Taking the orthogonal direct sum over cells proves the claim on the entire graph space, including one-state cells. □

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## References

[1] P. Caputo, *On the spectral gap of the Kac walk and other binary collision processes*, ALEA. Latin American Journal of Probability and Mathematical Statistics **4** (2008), 205–222. Primary article.

[2] E. Carlen, E. H. Lieb, and M. Loss, *An Inequality of Hadamard Type for Permanents*, Methods and Applications of Analysis **13** (2006), no. 1, 1–18. arXiv:math/0508096v1.

[3] C. J. Carstens, A. Berger, and G. Strona, *A unifying framework for fast randomization of ecological networks with fixed (node) degrees*, MethodsX **5** (2018), 773–780. doi:10.1016/j.mex.2018.06.018. Extended version: arXiv:1609.05137v3.

[4] C. J. Carstens and P. Kleer, *Speeding up Switch Markov Chains for Sampling Bipartite Graphs with Given Degree Sequence*, Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2018), Leibniz International Proceedings in Informatics **116** (2018), 36:1–36:18. doi:10.4230/LIPIcs.APPROX-RANDOM.2018.36. Preprint: *Comparing the Switch and Curveball Markov Chains for Sampling Binary Matrices with Fixed Marginals*, arXiv:1709.07290v3.

[5] C. Cooper, M. Dyer, and C. Greenhill, *Sampling regular graphs and a peer-to-peer network*, Combinatorics, Probability and Computing **16** (2007), no. 4, 557–593. doi:10.1017/S0963548306007978.

[6] C. Cooper, M. Dyer, and C. Greenhill, *Corrigendum: Sampling regular graphs and a peer-to-peer network*, preprint (2012), arXiv:1203.6111v1.

[7] P. Diaconis and M. Shahshahani, *Time to reach stationarity in the Bernoulli–Laplace diffusion model*, SIAM Journal on Mathematical Analysis **18** (1987), no. 1, 208–218. doi:10.1137/0518016.

[8] M. Dyer, C. Greenhill, and M. Ullrich, *Structure and eigenvalues of heat-bath Markov chains*, Linear Algebra and its Applications **454** (2014), 57–71. doi:10.1016/j.laa.2014.04.018.

[9] P. L. Erdős, C. Greenhill, T. R. Mezei, I. Miklós, D. Soltész, and L. Soukup, *The mixing time of switch Markov chains: a unified approach*, European Journal of Combinatorics **99** (2022), 103421. doi:10.1016/j.ejc.2021.103421.

[10] Y. Filmus, *An Orthogonal Basis for Functions over a Slice of the Boolean Hypercube*, Electronic Journal of Combinatorics **23** (2016), no. 1, P1.23. doi:10.37236/4567.

[11] W. Fu, Q. Qin, and G. Wang, *Spectral Gap for the Binary Fixed-Margin Swap Chain*, preprint (2026), arXiv:2606.22636v2.

[12] C. Greenhill and M. Sfragara, *The switch Markov chain for sampling irregular graphs and digraphs*, Theoretical Computer Science **719** (2018), 1–20. doi:10.1016/j.tcs.2017.11.010.

[13] V. Havel, *Poznámka o existenci konečných grafů* [A remark on the existence of finite graphs], Časopis pro pěstování matematiky **80** (1955), no. 4, 477–480. doi:10.21136/CPM.1955.108220.

[14] R. Kannan, P. Tetali, and S. Vempala, *Simple Markov-chain algorithms for generating bipartite graphs and tournaments*, Random Structures & Algorithms **14** (1999), no. 4, 293–308. doi:10.1002/(SICI)1098-2418(199907)14:4<293::AID-RSA1>3.0.CO;2-G.

[15] OpenAI, *Polynomial mixing of the switch chain for every graphical degree sequence*, OpenAI Math Release preprint OAI:Polynomial-Mixing-of-the-Switch-Chain-for-Every-Graphical-Degree-Sequence-September-25-2026, 2026.

[16] K. Tikhomirov and P. Youssef, *Sharp Poincaré and log-Sobolev inequalities for the switch chain on regular bipartite graphs*, Probability Theory and Related Fields **185** (2023), 89–184. doi:10.1007/s00440-022-01172-7.

[17] N. D. Verhelst, *An efficient MCMC algorithm to sample binary matrices with fixed marginals*, Psychometrika **73** (2008), no. 4, 705–728. doi:10.1007/s11336-008-9062-3.